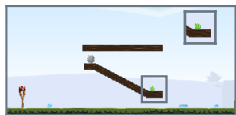


### (a) Observed state



Frozen CNN object slots



12 admissible launches



$a_i$

### (b) World-model prediction

Fixed request:  $\Delta=1$  model frame per step · continuous latent

One of 12 latent rollouts

launch  $a_i \rightarrow$  continuous state

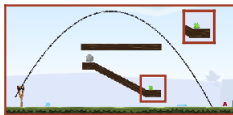
predicted task cost  $\hat{J}(a_i)$

Rank 12 predicted costs

Lowest predicted cost  $\rightarrow a_{09}$

### (c) Physics-engine outcome

**a09 · model's pick · miss**



**a06 · only success of 12 · target removed**



Independent engine replays of all 12 launches